MILITARY STANDARD

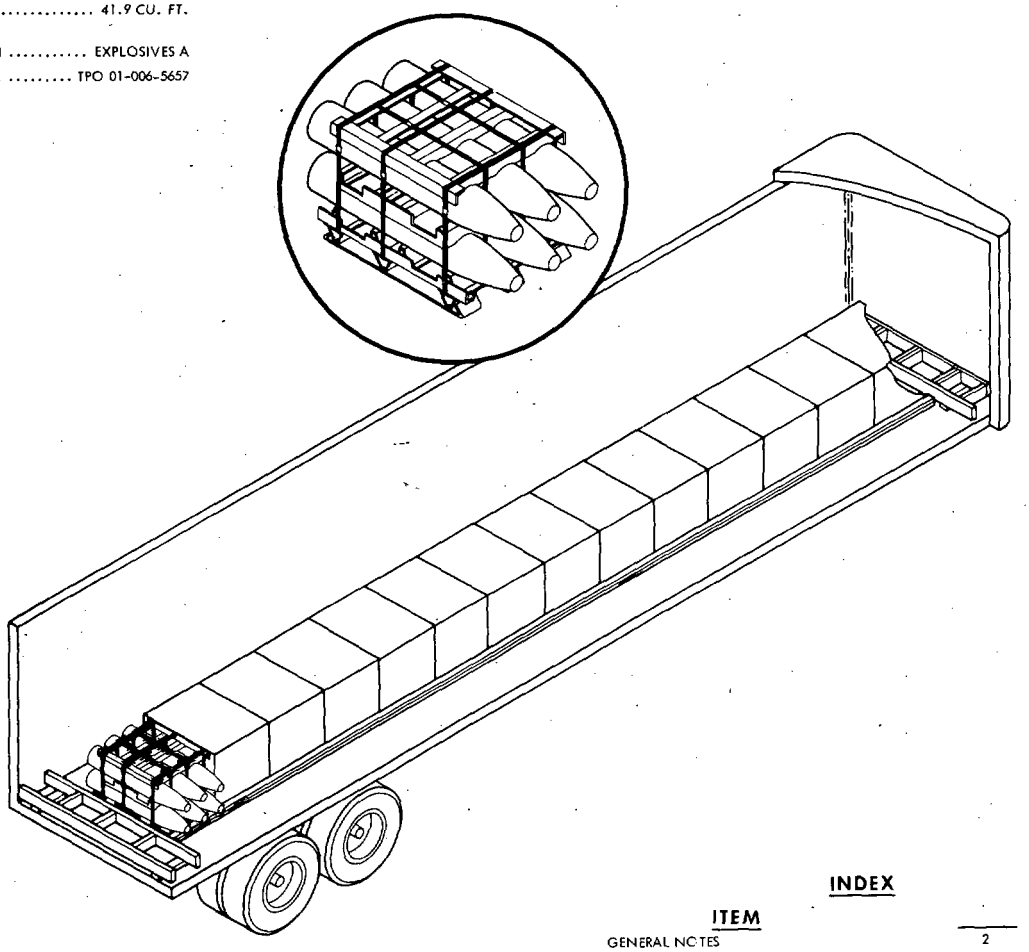
MIL-STD-1320-170
(NAVY)

12 JANUARY 1976

TRUCKLOADING
BOMB, MK 82 AND MODS (500 LBS)
IN MHU-149/E
AIR FORCE UNIT LOAD
TPO 01-006-5657

UNIT LOAD DATA

WEIGHT (APPROX) ..... 3035 LBS
DIMENSIONS ..... 61 3/4L X 35W X 33 1/2H
CUBE ..... 41.9 CU. FT.
DOT HAZARD CLASSIFICATION ..... EXPLOSIVES A
AIR FORCE DWG. NO. ..... TPO 01-006-5657



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS. REFER TO INDEX TO STANDARDS, MIL-HDBK-236. (NAVY).

INDEX

ITEM	
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FTL-SINGLE ROW 40 & 45 FT VAN	2 & 3
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14 UNIT LOADS	4 & 5
15 UNIT LOADS	6 & 7
FTL 40 & 42 FT FLATBED	8 & 9

AUTHORIZED FOR HIGHWAY MOVEMENT ONLY.

M.S. Gray NWHL 12/17/75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 12/17/75
SIGNATURE AIR SYSCOM, BY DIRECTION DATE

ORIGINATOR Charles MacBride 12/17/75
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING AIR FORCE UNIT LOADS OF 500 LB. MK 82 BOMBS WHEN THE BOMBS ARE PALLETIZED IN ACCORDANCE WITH TPO 01-006-5657.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 & 45 FT VANS AND 40 & 42 FT FLATBED TRAILERS. THE TRAILER MAY HAVE WOOD FLOORS, METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS AS SPECIFIED BY THE SPECIFIC PLAN. CHECK THE SPECIFIC LOAD PLAN FOR THE TYPE OF FLOOR PERMITTED. ALL TRAILERS SHALL HAVE THE TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. THE TRUCKLOAD SHOWN ON PAGE 7 IS FOR "LIGHTWEIGHT" TRACTOR-TRAILERS ONLY. THE EMPTY TRACTOR-TRAILER SHOULD NOT WEIGH OVER 25,500 LBS AND THE DRIVE AXLES SHOULD NOT EXCEED 10,500 LBS. IF THIS LOAD PLAN IS USED FOR A "STANDARD WEIGHT" TRACTOR-TRAILER (27,500 LBS NET VEHICLE WEIGHT), THE AXLE WEIGHT WILL BE IN EXCESS OF THE LIMIT OF 32,000 LBS. FOR TANDEM AXLES AND THE GROSS VEHICLE WEIGHT WILL EXCEED 73,280 LBS.
4. THE TRUCKLOADS SHOWN ON PAGES 3 AND 5 ARE SUITABLE FOR EITHER "STANDARD OR LIGHTWEIGHT" TRACTOR-TRAILERS.
5. THE FILLER ASSEMBLIES SHALL FILL THE VOID BETWEEN THE UNIT LOADS AND THE TRAILER SIDE WALLS. THEY SHALL BE SECURED IN PLACE TO PREVENT LONGITUDINAL MOVEMENT.

WARNING

THE USE OF FILLER ASSEMBLIES WHEN SHIPPING THE HEAVIER FTL'S MAY CAUSE THE VEHICLE TO BE OVER-WEIGHT. GROSS VEHICLE WEIGHTS AND AXLE WEIGHTS SHOULD BE COMPUTED PRIOR TO LOADING.

6. THE REAR BLOCKING ASSEMBLY SHALL BEAR AGAINST THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.

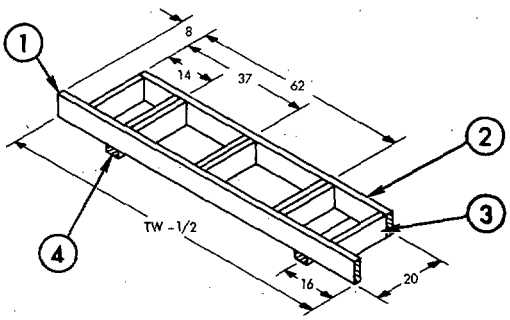
WARNING

DO NOT USE TRAILERS WITH ROLL UP DOORS WHEN BLOCKING BY THIS METHOD.

7. CHAINS AND STEEL STRAPS MAY BE USED INTERCHANGEABLY FOR TIE-DOWNS ON A 1 TO 1 BASIS.
8. STEEL STRAPS SHALL BE 2 X .050 AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENTS SHOWN ON PAGE 8.
9. CHAINS AND LOAD BINDERS SHALL MEET THE FOLLOWING CONDITIONS:
   A. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY.
   B. CHAINS, FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   C. LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
10. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.
11. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
12. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO TRAILER IN AN ACCESSIBLE AREA, AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
13. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I, CLASS A.
14. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL-SINGLE ROW (44 & 45 FT VANS)

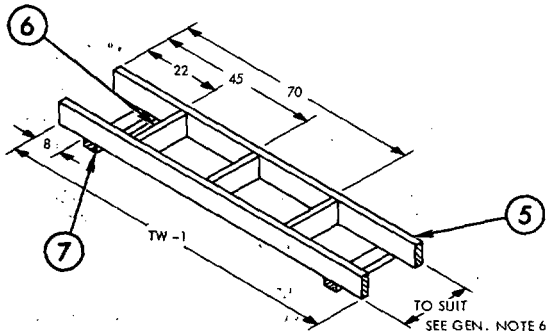
1. THIS TRUCKLOAD IS AUTHORIZED FOR A FTL (14 UNIT LOADS) ARRANGED IN A SINGLE ROW PATTERN. EITHER "STANDARD OR LIGHTWEIGHT" VANS HAVING FLOORS OF WOOD, METAL OR METAL WITH WOOD NAILING STRIPS MAY BE USED.
2. ONLY 44 & 45 FT VANS WILL ACCOMMODATE THE FTL OF 14 UNIT LOAD ARRANGED IN A SINGLE ROW PATTERN. WHEN 44 OR 45 FT VANS ARE NOT AVAILABLE, USE THE TRUCKLOAD SHOWN ON PAGE 5 TO SHIP 14 UNIT LOADS.
3. THE FRONT BULKHEAD, DETAIL A, IS FOR 44 FT TRAILERS. WHEN LOADING 45 FT TRAILERS, INCREASE THE DEPTH OF THE BULKHEAD FROM 20 TO 26 INCHES.
4. DETAIL B MAY BE OF "SOLID FILL" CONSTRUCTION WHEN SPACE AT REAR OF LOAD IS LESS THAN 12 INCHES.
5. IF THE TRAILER HAS ALL METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS THAT ARE POSITIONED WRONG FOR NAILING, USE FILLER ASSEMBLY DETAIL D, PAGE 4, TO CONTROL LATERAL MOVEMENT OF LADING.



DETAIL A

FRONT BULKHEAD (44 FT TRAILER)
(SEE FTL NOTE 3, PAGE 2)
(1 REQUIRED)

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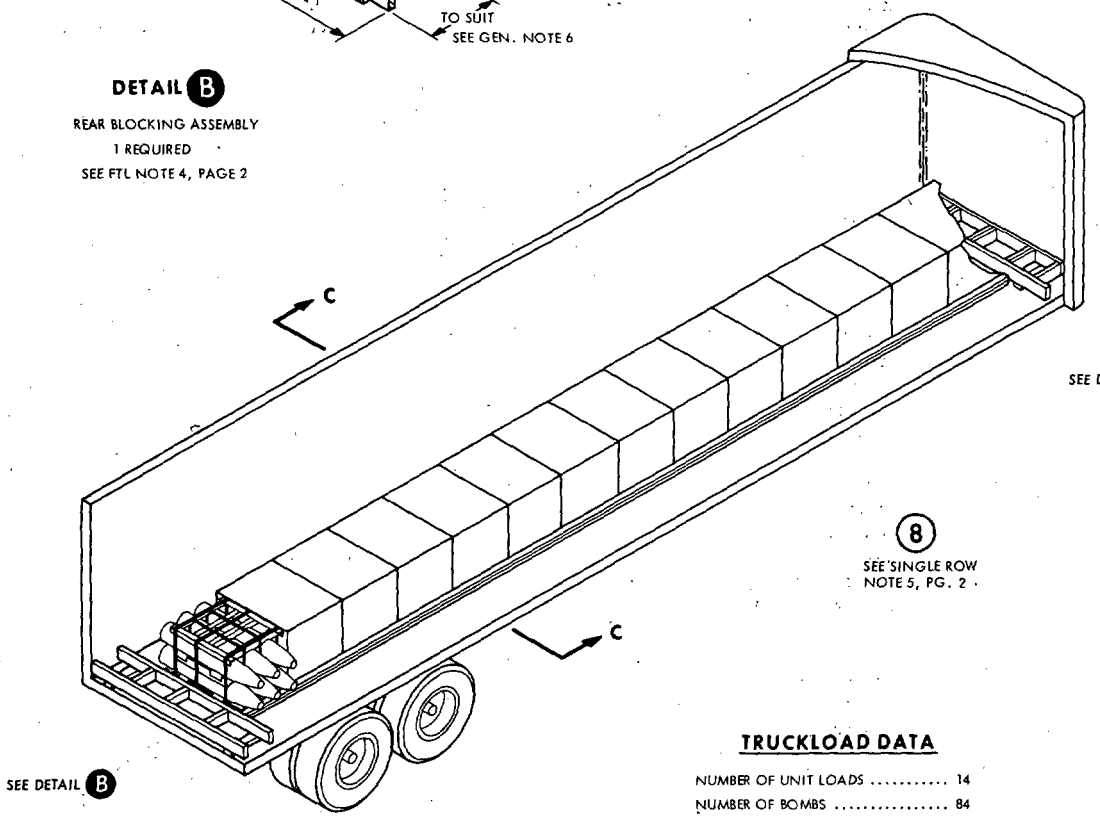


DETAIL B

REAR BLOCKING ASSEMBLY
1 REQUIRED
SEE FTL NOTE 4, PAGE 2

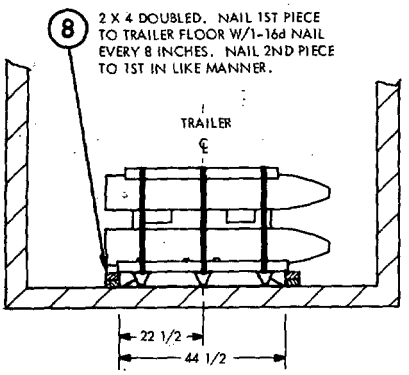
NOTE

THE TRUCKLOAD SHOWN BELOW IS FOR A 44 FT VAN WITH A WOOD FLOOR. FOR 45 FT VANS SEE FTL NOTE 3. FOR VANS WITH NON-NAILABLE FLOORS, SEE FTL NOTE 5.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 14
NUMBER OF BOMBS 84
TRUCKLOAD WEIGHT ..... 42,490 LBS



SECTION C-C

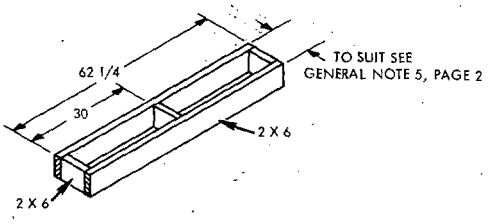
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
8	SLEEPER	2 X 4 X AS REQ'D	164 FT	SEE SECTION C-C		
7	SUPPORT	2 X 4 TO SUIT	2	5	2/JOINT	16d
6	STRUT	2 X 6 TO SUIT	3	SEE 5	-	-
5	CROSSMEMBER	2 X 6 X TW-1	2	6	3/JOINT	-
4	SUPPORT	2 X 4 X 20	2	1 & 2	2/JOINT	16d
3	STRUT	2 X 6 X 17	8	SEE 1 & 2	-	-
2	FWD CROSSMEMBER	2 X 6 TO SUIT	1	3	3/JOINT	16d
1	AFT CROSSMEMBER	2 X 6 X TW-1	1	3	3/JOINT	16d

LIST OF MATERIALS & NAILING DATA

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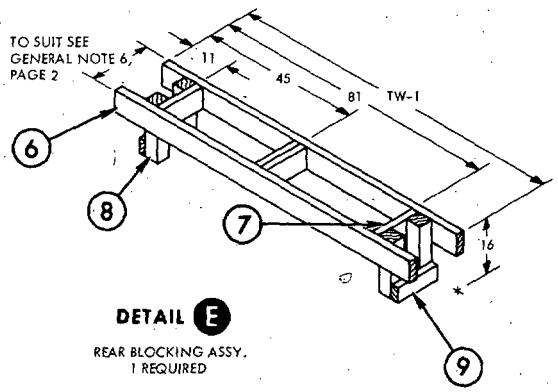
FTL-DOUBLE ROW PATTERN (40, 42, 44 & 45 FT. VANS) 14 UNIT LOADS

1. THIS TRUCKLOAD IS AUTHORIZED FOR A FTL (14 UNIT LOADS) ARRANGED IN A DOUBLE ROW PATTERN. EITHER "STANDARD OR LIGHT-WEIGHT" VANS HAVING FLOORS OF WOOD, METAL OR METAL WITH WOOD NAILING STRIPS MAY BE USED.
2. THE DEPTH OF THE FRONT BULKHEAD, DETAIL F, VARIES WITH THE LENGTH OF THE TRAILER BEING LOADED. THE BULKHEAD SHALL BE CONSTRUCTED AS SPECIFIED BY DETAIL F AND TABLE I.
3. IF THE TRAILER HAS ALL METAL FLOOR OR METAL FLOOR WITH WOOD NAILING STRIPS THAT ARE LOCATED WRONG FOR NAILING, USE FILLER ASSEMBLY DETAIL D TO CONTROL LATERAL MOVEMENT OF LADING.
4. WHEN SPACE AT THE REAR OF THE LOADED TRAILER PERMITS AND THE TRAILER HAS A NAILABLE FLOOR, DETAIL G IS AN AUTHORIZED ALTERNATE. TWO KNEE BRACES ARE REQUIRED FOR EACH UNIT LOAD (4 TOTAL), WITH THE BRACE AGAINST THE 1ST AND 3RD BOMBS OF THE BOTTOM LAYER OF THE UNIT LOAD. NAIL BRACE TO TRAILER FLOOR WITH 8-16d NAILS.



DETAIL D

FILLER ASSEMBLY
SEE NOTE 3, PAGE 4

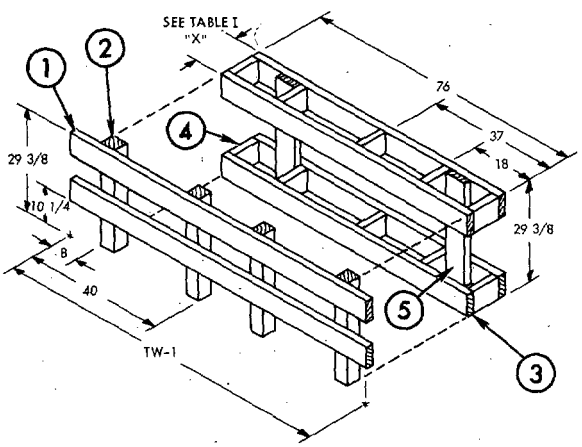


DETAIL E

REAR BLOCKING ASSY.
1 REQUIRED

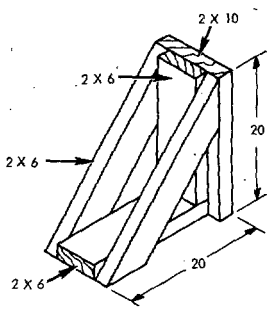
TABLE I

LENGTH OF TRAILER	DIMENSION "X"
40 FT	12
42 FT	25
44 FT	37
45 FT	43



DETAIL F

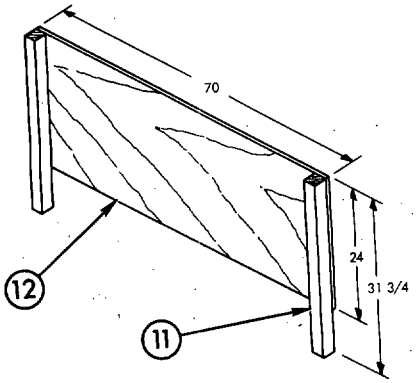
FRONT BULKHEAD
1 REQUIRED
SEE NOTE 2, PAGE 4



DETAIL G

KNEE BRACE
ALTERNATE REAR BLOCKING
SEE NOTE 4, ABOVE

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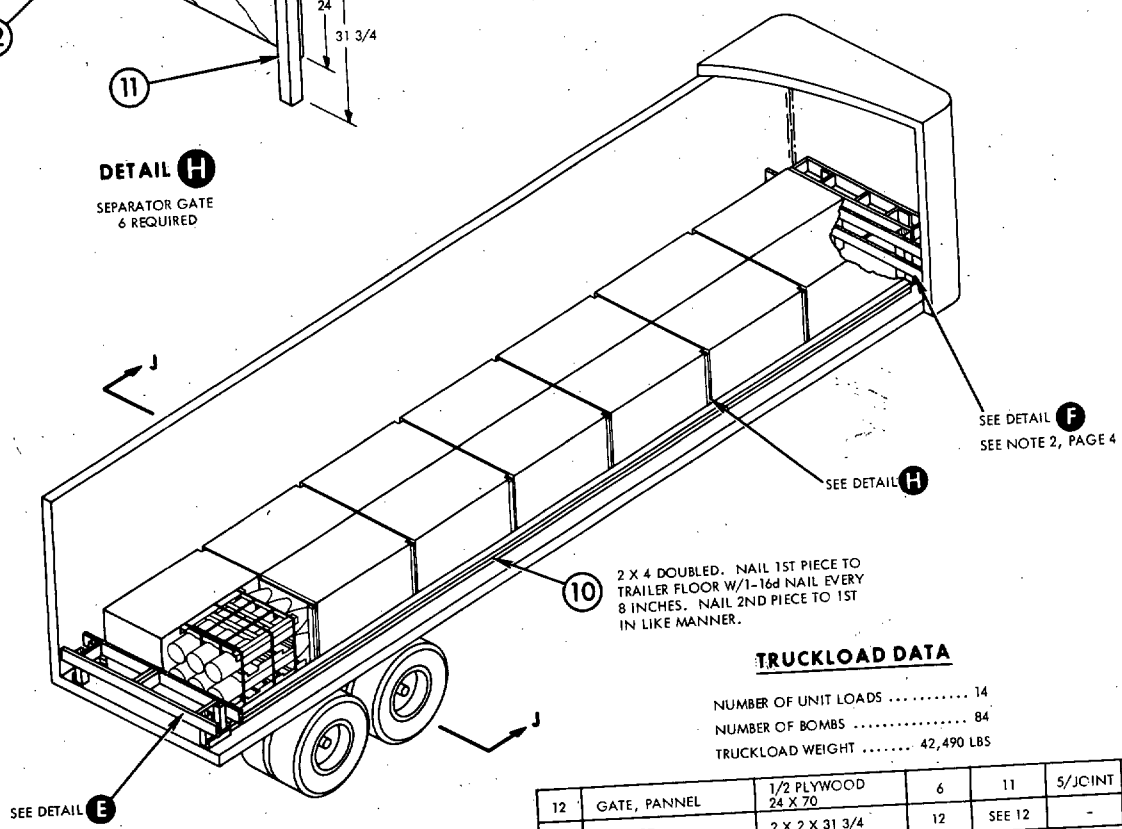


DETAIL H

SEPARATOR GATE
6 REQUIRED

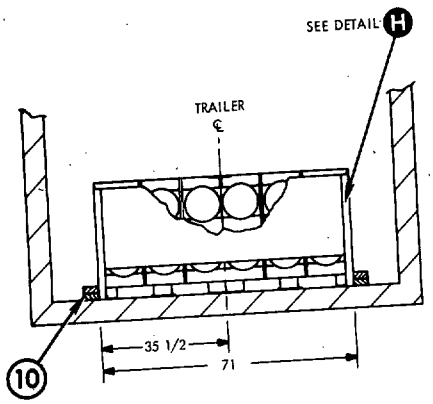
NOTE

THE TRUCKLOAD SHOWN BELOW IS FOR VANS WITH WOOD FLOORS. FOR VANS WITH NON-NAILABLE FLOORS SEE NOTE 3, PAGE 4.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 14
NUMBER OF BOMBS 84
TRUCKLOAD WEIGHT ..... 42,490 LBS



SECTION J-J

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
LIST OF MATERIALS & NAILING DATA						
12	GATE, PANNEL	1/2 PLYWOOD 24 X 70	6	11	5/JOINT	6d
11	SUPPORT	2 X 2 X 31 3/4	12	SEE 12	-	-
10	SLEEPER	2 X 4 X AS REQ'D	146 FT	SEE FIELD NOTE		
9	TIE	2 X 4 TO SUIT	2	8	2/JOINT	12d
8	LEG	4 X 4 X 6	4	SEE 6 & 7	-	-
7	STRUT	2 X 6 TO SUIT	4	SEE 6 8	- 2/JOINT	- 16d
6	CROSSMEMBER	2 X 6 X TW-1	2	7 8	3/JOINT 2/JOINT	16d 16d
5	SUPPORT	2 X 6 X 29 3/8	2	3	3/JOINT	16d CLIN.
4	STRUT	2 X 6 TO SUIT	10	SEE 3	-	-
3	CROSSMEMBER	2 X 6 X 76	4	4 & 3	3/JOINT	16d
2	VERTICAL	4 X 4 X 29 3/8	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW-1	2	2	3/JOINT	16d

FTL-DBL. ROW (14 UNIT LOAD)

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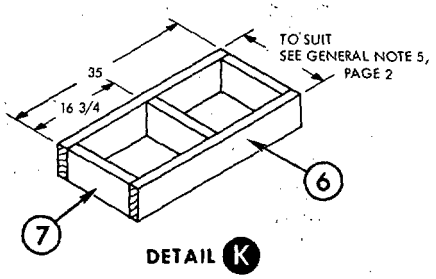
FTL-DBL ROW + 1 PATTERN (42, 44 & 45 FT. VANS) 15 UNIT LOADS

1. THIS TRUCKLOAD IS AUTHORIZED FOR A FTL (15 UNIT LOADS) ARRANGED IN A DOUBLE ROW PATTERN PLUS ONE UNIT LOAD CROSSWAYS AT THE REAR OF THE LOAD. ONLY LIGHTWEIGHT VANS HAVING FLOORS OF WOOD, METAL OR METAL WITH WOOD NAILING STRIPS MAY BE USED.

WARNING

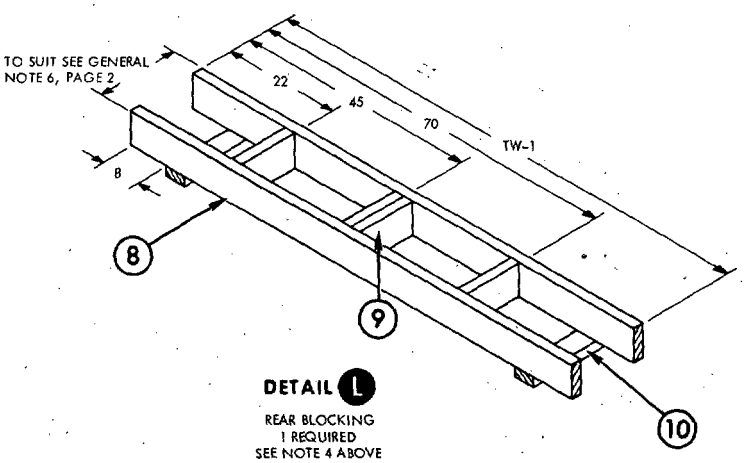
THE EMPTY WEIGHT OF "LIGHTWEIGHT" TRACTOR-TRAILERS SHOULD NOT EXCEED 25,500 LBS OR THE DRIVE AXLES 10,500 LBS.

2. THE DEPTH OF THE FRONT BULKHEAD, DETAIL M, VARIES WITH THE LENGTH OF THE TRAILER BEING LOADED. THE BULKHEAD SHALL BE CONSTRUCTED AS SPECIFIED BY DETAIL M AND TABLE II.
3. IF THE TRAILER HAS ALL METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS THAT ARE LOCATED WRONG FOR NAILING, USE FILLER ASSEMBLY, DETAIL D, ON PAGE 4 TO CONTROL LATERAL MOTION. SEE GENERAL NOTE 5 PAGE 2.
4. DETAIL L MAY BE OF "SOLID FILL" CONSTRUCTION WHEN SPACE AT REAR OF LOAD IS LESS THAN 12 INCHES.



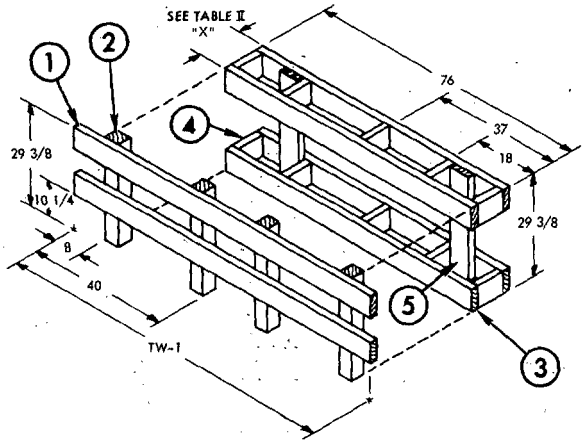
DETAIL K

FILLER ASSEMBLY
2 REQUIRED



DETAIL L

REAR BLOCKING
1 REQUIRED
SEE NOTE 4 ABOVE



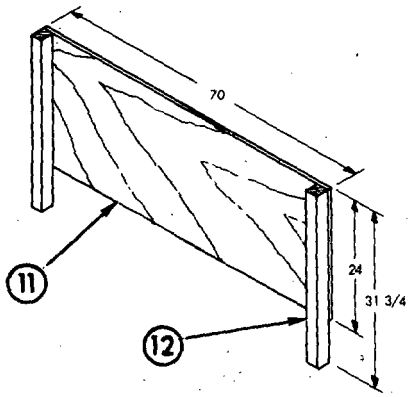
DETAIL M

FRONT BULKHEAD
1 REQUIRED
SEE NOTE 2 ABOVE

TABLE II

LENGTH OF TRAILER	DIMENSION "X"
42	12
44	24
45	30

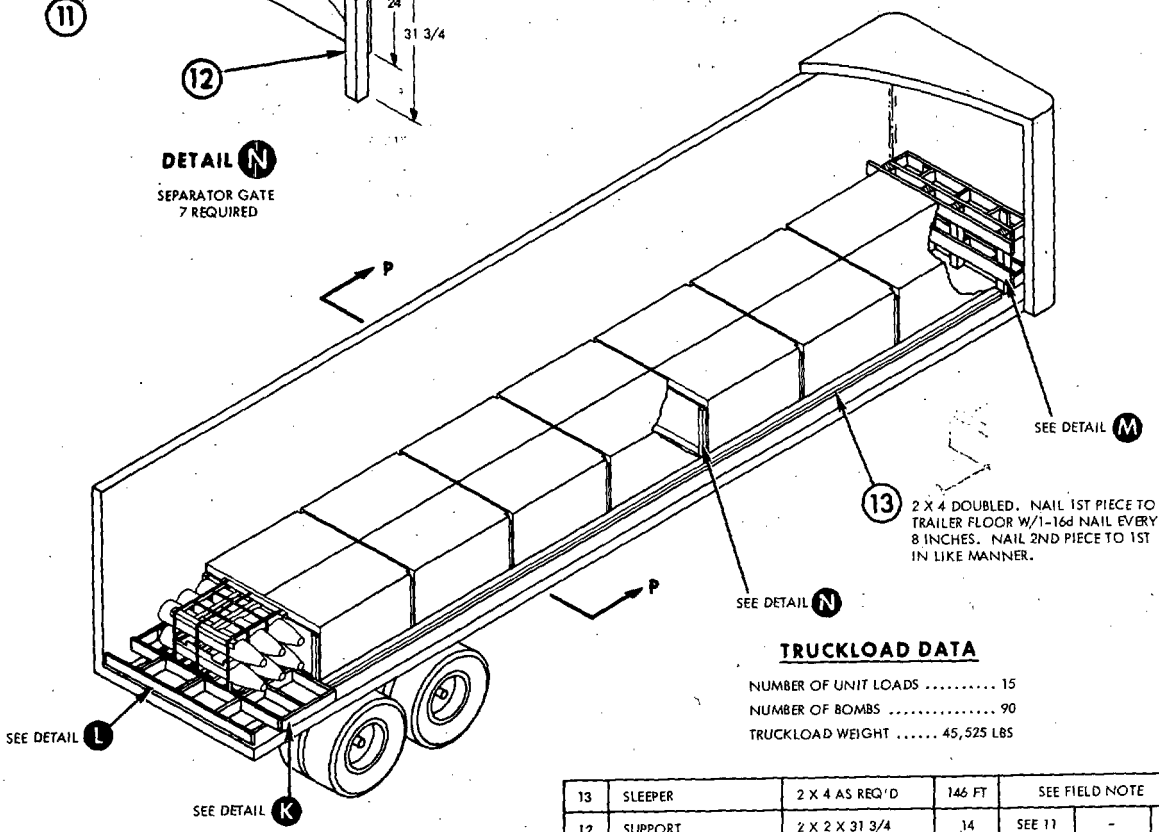
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DETAIL N
SEPARATOR GATE
7 REQUIRED

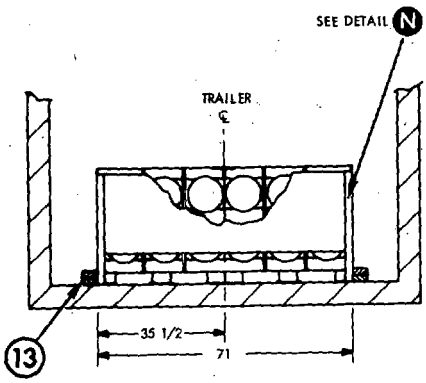
NOTE

THE TRUCKLOAD SHOWN BELOW IS FOR VANS WITH NAILABLE FLOORS. IF VAN FLOORS ARE NOT NAILABLE SEE NOTE 3 PAGE 6.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 15
NUMBER OF BOMBS 90
TRUCKLOAD WEIGHT ..... 45,525 LBS



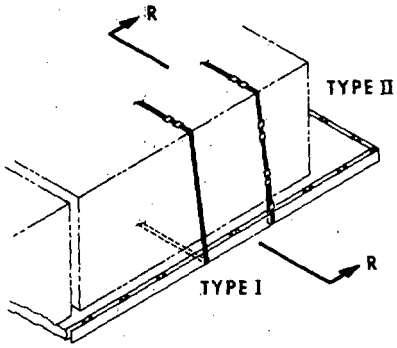
SECTION P-P

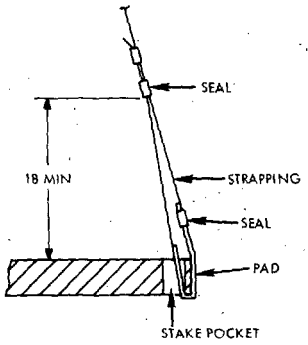
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
13	SLEEPER	2 X 4 AS REQ'D	146 FT	SEE FIELD NOTE		
12	SUPPORT	2 X 2 X 31 3/4	14	SEE 11	-	-
11	GATE PANNEL	1/2 PLYWOOD 24 X 70	7	12	5/JOINT	6d
10	SUPPORT	2 X 4 TO SUIT	2	8	2/JOINT	12d
9	STRUT	2 X 6 TO SUIT	4	SEE 8	-	-
8	CROSSMEMBER	2 X 6 X TW-1	2	9	3/JOINT	16d
7	STRUT	2 X 6 TO SUIT	6	SEE 6	-	-
6	LONGITUDINAL	2 X 6 X 35	4	7	3/JOINT	16d
5	SUPPORT	2 X 6 X 29 3/8	2	3	3/JOINT	16d CLIN.
4	STRUT	2 X 6 TO SUIT	10	SEE 3	-	-
3	CROSSMEMBER	2 X 6 X 76	4	4 & 3	3/JOINT	16d
2	VERTICAL	4 X 4 X 29 3/8	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW-1	2	2	3/JOINT	16d

LIST OF MATERIALS & NAILING DATA

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LOAD STRAPPING





SECTION R-R

TIE-DOWN STRAPPING

THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAPPING SEALS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PER CENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION, AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD, IF PRACTICABLE.

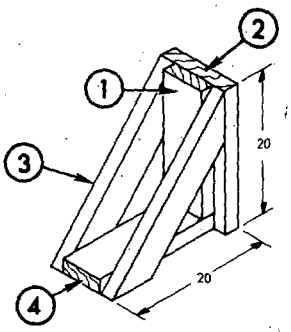
TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD.

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH TWO DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION R-R. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STACK POCKET PAD (A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

FTL-DOUBLE ROW PATTERN (40 & 42 FT FLATBED TRAILER)

14 UNIT LOADS

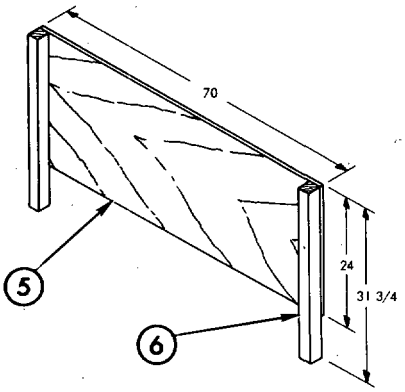
1. THIS TRUCKLOAD IS AUTHORIZED FOR A TFL (14 UNIT LOADS) ARRANGED IN A DOUBLE ROW PATTERN.
2. THE LADING SHALL BE COVERED WITH A WATERPROOF, FIRE RESISTANT TARPAULIN. THE BEST RESULTS ARE OBTAINED IF THE UNIT LOADS ARE COVERED WITH THE TARPAULIN PRIOR TO APPLYING THE TIE-DOWN CHAINS OR STRAPS. PROTECTOR BOARDS SHALL BE USED BETWEEN THE TIE-DOWNS AND THE TARPAULIN.
3. WHEN LOADING 42 FT FLATBED TRAILERS, POSITION THE 1ST UNIT LOADS 34 INCHES FROM FRONT OF TRAILER.
4. POSITION PROTECTOR BOARD, PIECE 7, AND HOLD DOWN CHAIN OR STRAPS OVER THE "HARD POINTS" OF THE UNIT LOADS.
5. CHAIN AND 2 INCH STRAPS MAY BE USED INTERCHANGEABLY ON A 1 TO 1 BASIS.
6. STEEL STRAPS SHALL BE 2 X .050 AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENT SHOWN ABOVE.



DETAIL S

KNEE BRACE
12 REQUIRED

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DETAIL T

SEPARATOR GATE
6 REQUIRED

NOTE

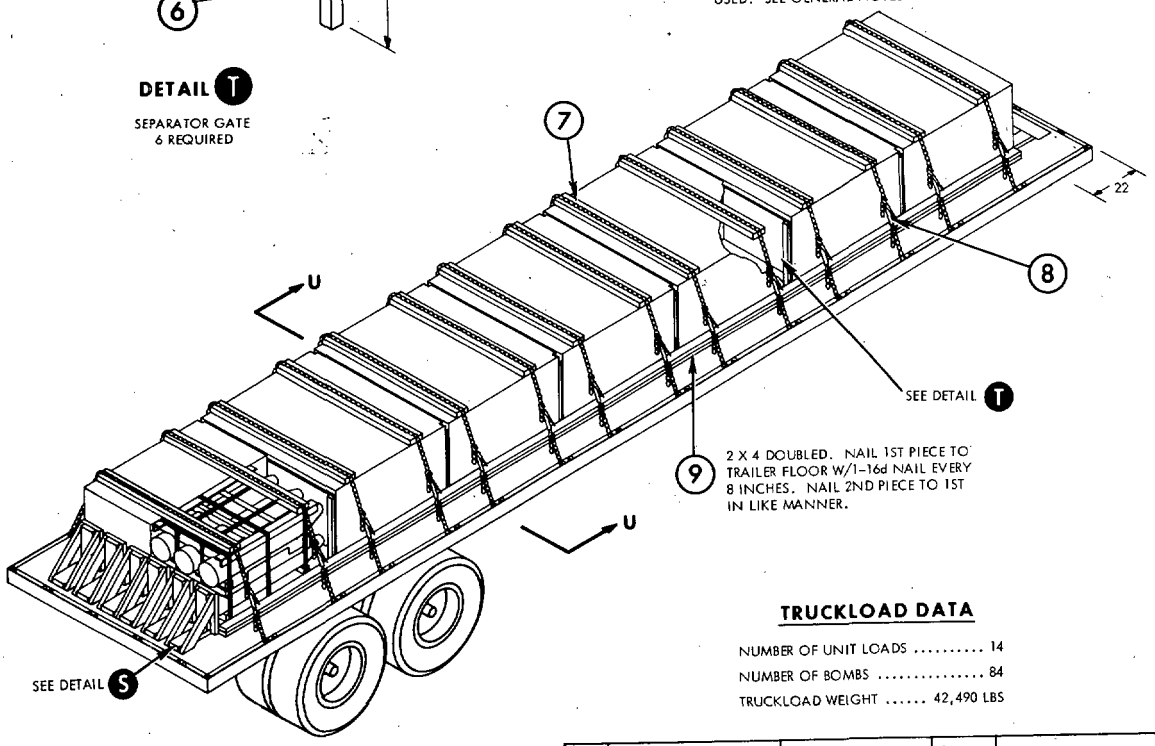
TARPAULIN OMITTED FOR CLARITY.

NOTE

LOAD ALL UNITS LOADS WITH BOMB NOSES FORWARD.

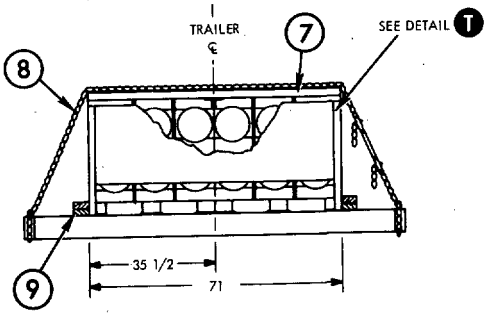
NOTE

THIS LOAD ILLUSTRATES THE USE OF CHAINS AS TIEDOWNS. STEEL STRAPS MAY ALSO BE USED. SEE GENERAL NOTES 7 & 8.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 14
NUMBER OF BOMBS 84
TRUCKLOAD WEIGHT ..... 42,490 LBS



SECTION U-U

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
9	SLEEPER	2 X 4 AS REQ'D	146 FT	SEE FIELD NOTE		
8	CHAIN & LOADBINDER	3/8 CAPACITY	14	-	-	-
7	PROTECTOR BOARD	2 X 6 X 70	14	-	-	-
6	SUPPORT	2 X 2 X 31 3/4	12	SEE 5	-	-
5	GATE PANNEL	1/2 PLYWOOD 24 X 70	6	6	5/JOINT	6d
4	BRACE, CLEAT	2 X 6 X 18 1/2	12	TRAILER FLOOR	6	16d
3	BRACE DIAGONAL	2 X 6 TO SUIT	24	1 & 4	4/JOINT	16d
2	BRACE BACKUP	2 X 10 X 20	12	4	4	16d
1	BRACE UPRIGHT	2 X 6 X 18 1/2	12	2	6	16d

LIST OF MATERIALS & NAILING DATA

FTL-40 & 42 FT FLATBED TRAILER

REVIEW ACTIVITY
NAVY: OS, AS

PREPARING ACTIVITY
NAVY: OS
(PROJECT NUMBER 8140-N307)

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